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Network Security and Cryptography Assignment Solutions

Subject: EE-482: Network Security and Cryptography
Session: 2021, Spring-2025 Semester
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Question 1: Security Attacks [10 Marks]

Definition of Security Attacks

Security attacks are deliberate attempts to compromise the security of a computer system, network, or data. These attacks aim to violate one or more security principles such as confidentiality, integrity, availability, authenticity, or non-repudiation.

Distinction Between Passive and Active Attacks

Passive Attacks

- **Definition:** Attacks that attempt to learn or make use of information from the system without affecting system resources
- **Characteristics:**
  - Do not modify data or system resources
  - Difficult to detect because they don't alter the data
  - Focus on eavesdropping and monitoring transmissions
  - Violate confidentiality

Examples of Passive Attacks:

1. **Eavesdropping/Interception:** Monitoring network traffic to capture sensitive information
2. **Traffic Analysis:** Analyzing communication patterns to gather intelligence
3. **Packet Sniffing:** Capturing data packets transmitted over a network
4. **Shoulder Surfing:** Observing someone entering passwords or sensitive information

Active Attacks

- **Definition:** Attacks that attempt to alter system resources or affect their operation
- **Characteristics:**
  - Modify data or system resources

- Easier to detect but harder to prevent
- Can disrupt system operations
- Violate integrity, availability, and authenticity

Examples of Active Attacks:

1. **Masquerading/Spoofing:** Pretending to be a legitimate user or system
2. **Replay Attack:** Retransmitting captured data to gain unauthorized access
3. **Modification of Messages:** Altering data in transit
4. **Denial of Service (DoS):** Overwhelming system resources to make them unavailable
5. **Man-in-the-Middle Attack:** Intercepting and potentially altering communications

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Question 2: Significant Features of Network Security [10 Marks]

Network security encompasses several key features that work together to protect networked systems and data:

1. Confidentiality

- Ensures that information is accessible only to authorized parties
- Prevents unauthorized disclosure of sensitive data
- Implemented through encryption, access controls, and secure communication channels

2. Integrity

- Ensures that data has not been altered or corrupted during transmission or storage
- Maintains accuracy and completeness of information
- Achieved through checksums, digital signatures, and hash functions

3. Availability

- Ensures that network resources and services are accessible when needed
- Prevents disruption of services and system downtime
- Maintained through redundancy, load balancing, and DDoS protection

4. Authentication

- Verifies the identity of users, devices, or systems
- Ensures that entities are who they claim to be

- Implemented through passwords, biometrics, certificates, and multi-factor authentication

5. Authorization

- Controls access to resources based on authenticated identity
- Defines what actions users can perform on specific resources
- Managed through access control lists (ACLs) and role-based access control (RBAC)

6. Non-repudiation

- Prevents denial of actions performed by authenticated users
- Provides proof of origin and delivery of messages
- Achieved through digital signatures and audit trails

7. Accountability

- Tracks and logs user activities for auditing purposes
- Enables detection of security violations and forensic analysis
- Maintained through comprehensive logging and monitoring systems

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Question 3: Repudiation and Prevention [10 Marks]

What is Repudiation?

Repudiation is the denial of participation in a communication or transaction by one of the parties involved. It occurs when someone denies having sent a message, received a message, or performed a particular action, even though they actually did.

Types of Repudiation:

1. **Repudiation of Origin:** Sender denies sending a message
2. **Repudiation of Delivery:** Receiver denies receiving a message

Real-Life Example (Check Scenario):

When you issue a check and the bank debits your account, if you later claim you never issued that check, this is repudiation. The bank needs proof that you actually issued the check to prevent fraudulent claims.

Prevention Methods:

1. Digital Signatures

- Uses cryptographic techniques to bind the sender's identity to the message

- Provides proof of origin and integrity
- Cannot be forged without the sender's private key

2. Timestamps

- Records the exact time when an action was performed
- Prevents replay attacks and establishes sequence of events
- Often combined with digital signatures for enhanced security

3. Audit Trails

- Maintains detailed logs of all transactions and communications
- Provides evidence of activities performed by users
- Enables forensic analysis and dispute resolution

4. Certificates and Public Key Infrastructure (PKI)

- Establishes trust through third-party certificate authorities
- Binds public keys to identities
- Enables verification of digital signatures

5. Secure Hash Functions

- Creates unique fingerprints of messages
- Detects any modifications to the original content
- Used in conjunction with digital signatures

6. Multi-factor Authentication

- Requires multiple forms of verification
- Makes it harder for attackers to impersonate legitimate users
- Strengthens the authentication process

Question 4: Playfair Cipher Encryption/Decryption [10 Marks]

Playfair Cipher for "FREEDOM" with key "POSSIBLE"

Step 1: Prepare the Key Matrix

Key: POSSIBLE
Remove duplicates: POSIBLE

Complete alphabet: POSIBLEACDFGHJKMNQRTUVWXYZ

5x5 Key Matrix:

```
P O S I B
L E A C D
F G H J K
M N Q R T
U V W X Y
```

Step 2: Prepare the Plaintext

Plaintext: FREEDOM

- Group into pairs: FR EE DO M
- Add filler 'X' for odd length: FR EE DO MX
- Handle double letters in EE: FR EX DO MX

Plaintext pairs: FR, EX, DO, MX

Step 3: Encryption Process

Pair 1: FR

- F is at position (3,1), R is at position (4,4)
- Different row, different column → Rectangle rule
- F → (3,4) = J, R → (4,1) = M
- **FR → JM**

Pair 2: EX

- E is at position (2,2), X is at position (5,4)
- Different row, different column → Rectangle rule
- E → (2,4) = C, X → (5,2) = V
- **EX → CV**

Pair 3: DO

- D is at position (2,5), O is at position (1,2)
- Different row, different column → Rectangle rule
- D → (2,2) = E, O → (1,5) = B

- **DO → EB**

Pair 4: MX

- M is at position (4,1), X is at position (5,4)
- Different row, different column → Rectangle rule
- M → (4,4) = R, X → (5,1) = U
- **MX → RU**

Final Result:

Ciphertext: JMCVEBRU

Decryption Process:

To decrypt, reverse the process using the same rules:

- JMCVEBRU → JM CV EB RU
- Apply reverse Playfair rules to get: FR EX DO MX
- Remove filler and adjust: FREEDOM

Question 5: Diffie-Hellman Key Exchange [10 Marks]

Given Parameters:

- p = 11 (prime modulus)
- $\alpha = 7$ (generator)

Solution:

Step 1: Choose Private Keys (Assumptions)

- **$X_A = 3$** (Alice's private key)
- **$X_B = 6$** (Bob's private key)

Step 2: Calculate Public Keys

$Y_A = \alpha^{X_A} \bmod p$ $Y_A = 7^3 \bmod 11 = 343 \bmod 11 = 2$

$Y_B = \alpha^{X_B} \bmod p$ $Y_B = 7^6 \bmod 11 = 117649 \bmod 11 = 4$

Step 3: Calculate Shared Secret Key

Alice calculates K: K = Y_B^X_A mod p = 4^3 mod 11 = 64 mod 11 = 9

Bob calculates K: K = Y_A^X_B mod p = 2^6 mod 11 = 64 mod 11 = 9

Final Values:

- **X_A = 3** (Alice's private key)
- **X_B = 6** (Bob's private key)
- **Y_A = 2** (Alice's public key)
- **Y_B = 4** (Bob's public key)
- **K = 9** (Shared secret key)

Verification:

Both parties successfully derived the same shared secret key K = 9, demonstrating successful key exchange.

Process Summary:

1. Alice and Bob agree on public parameters p = 11 and α = 7
2. Each chooses a private key (X_A, X_B)
3. Each calculates and shares their public key (Y_A, Y_B)
4. Each uses the other's public key with their own private key to calculate the shared secret K
5. Both arrive at the same shared secret K = 9

Note: This solution demonstrates the mathematical principles of each cryptographic concept. In practice, these implementations would require additional security considerations and proper implementation details.